Unified Engineering Workbench

Design, implementation and verification report · Windows x64 prototype

1. Design overview

The prototype integrates geometry import, steady 1D thermal finite elements, parallel batch solves and embedded-system representation in one local workbench. A native C++17 Windows launcher starts a loopback-only service and opens the bundled responsive browser interface. All parsing, numerical computation, workload execution and mock adapter behavior run in C++; the frontend handles file selection, interaction, plots and export.

Five views cover Geometry, Thermal Analysis, Performance, Hardware and Backpropagation. The user can inspect ASCII STL triangles and temperature nodes in 3D; solve/export the rod case; run repeatable CPU measurements; and exercise simulated device discovery, connection, telemetry and timeout. Input errors name malformed headers, empty files, invalid/non-finite values and the 20 MiB limit. Browser requests are asynchronous so the interface stays responsive during computation.

Architecture and delivery

Layer	Responsibility
C++ core	STL/CSV parsing; assembled FEM and tridiagonal solve; fixed-seed cases and bounded pool; explicit gradients and finite-difference check; device mocks.
Launcher	Windows entry point, localhost HTTP transport, default-browser launch, occupied-port message and orderly close. Static GCC runtime components are linked in the packaged build.
Browser UI	Responsive controls, 3D inspection, charts, status, validation messages and CSV exports. It does not duplicate backend numerical or file-validation logic.

The folder includes C++ source, CMake project, editor run configuration, synthetic data with provenance, raw results, verification script, dependency record, readable Git log/bundle, AI-use disclosure and this five-page report.

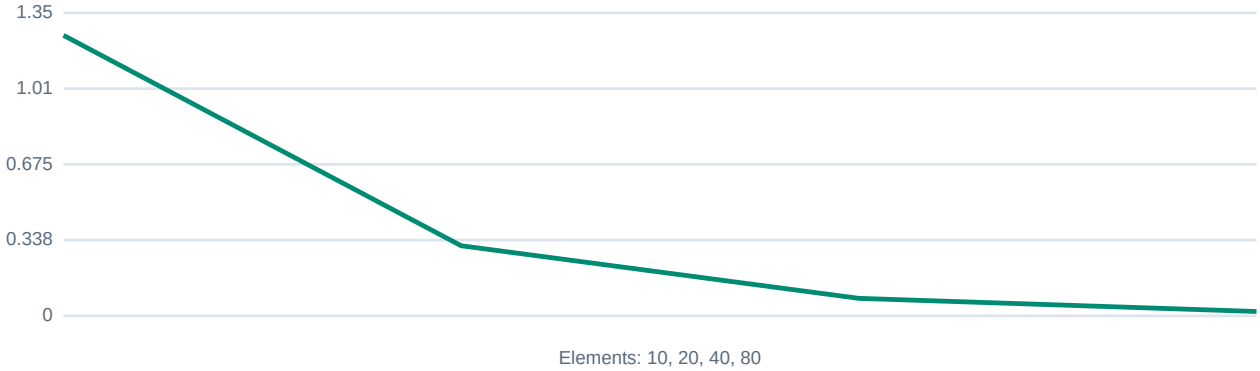
GitHub repository URL: https://github.com/shivanshOH/RA2311026010197-engineering-workbench

2. Thermal finite-element verification

The steady rod equation is -d/dx(kA dT/dx)=qA with constant properties and insulated lateral surfaces. A two-node element has K(e)=(kA/he)[[1,-1],[-1,1]] and f(e)=qAhe/2[1,1]^T. Assembly yields a tridiagonal system. Both prescribed end temperatures are imposed by eliminating known values into the right-hand side and replacing boundary rows; the reduced system is solved with the Thomas algorithm.

The case uses L=1 m, A=1e-4 m², k=10 W/(m K), q=10,000 W/m³, T(0)=100 °C and T(L)=20 °C. The exact solution is T_exact=100-80x+500x(1-x) °C. Maximum absolute error is measured at element midpoints from FE linear interpolation.

Mesh convergence: maximum midpoint error



Elements	Midpoint error (°C)	Reduction
10	1.250000	—
20	0.312500	4.00x
40	0.078125	4.00x
80	0.019531	4.00x

Halving the element size reduces error by four, consistent with second-order convergence. The exact quadratic and FE solution agree at the nodes; midpoint error reveals the curvature that linear interpolation does not capture. A debugging check was the source-term sign: positive generation must bend interior temperature above the end-to-end linear profile. The underlying nodal CSV files for all four meshes are included in results/.

Assumptions exclude temperature-dependent material properties, lateral heat loss, contact resistance and transient effects. Coordinates are metres; temperatures are degrees Celsius.

3. Multicore measurements

Independent cases use seed 20261026, k uniform in [5,25] W/(m K) and q uniform in [1,000,20,000] W/m³, with 40 elements per case. Serial and parallel versions use the same case sequence and solver, and write results in deterministic input order. Worker counts 1–4 are capped by available logical processors. The first run warms up; the following five wall-clock runs are summarized by median. File selection, loading, rendering and compilation are excluded; pool construction and joining are included.

Workers	Median ms	Speedup	Efficiency	Max abs difference (°C)
1	184.110	1.000x	100.0%	0
2	136.875	1.345x	67.3%	0
3	112.573	1.635x	54.5%	0
4	138.559	1.329x	33.2%	0

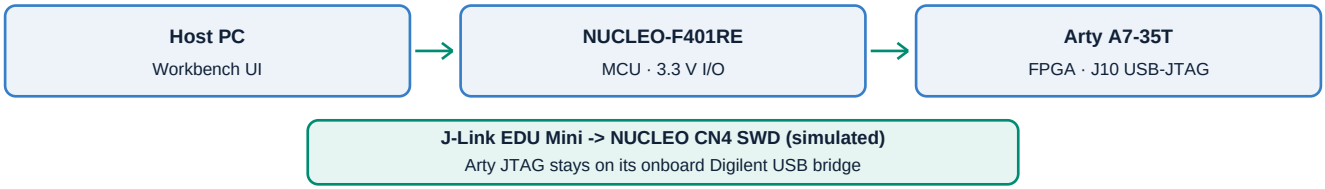
Observed test machine

CPU	12th Gen Intel(R) Core(TM) i7-12700H
Physical / logical cores	14 / 20
Installed RAM	15.6 GiB reported
Compiler and flags	GCC 16.2.0, C++17, -O2, statically linked libgcc/libstdc++
Workload	50,000 cases x 40 elements; 1 warm-up + 5 repetitions; median
Agreement	0 °C displayed maximum difference; required ≤ 1e-8 °C
Launch-to-ready	796.8 ms to first local ready signal, observed once

Three workers were fastest in this observed run (1.64x). Four workers were slower than three and only moderately faster than serial. Scheduling, pool creation, memory pressure and case granularity limit scaling; this is one laptop measurement, not a general guarantee. Five raw timing repetitions per worker and metadata are in results/performance_timings.csv.

4. Embedded integration design

The documented device set is an ST NUCLEO-F401RE MCU board, Digilent Arty A7-35T FPGA board and SEGGER J-Link EDU Mini debug adapter. The host connects independently to NUCLEO USB/ST-LINK virtual COM and Arty J10 onboard USB-JTAG. The J-Link targets the MCU over SWD at CN4. The adapter transports debug signals; it is not a compute board or a generic FPGA programmer. All behaviors shown in the application are simulated.



Signal/link	Role and assumption	Domain
CN4-1 VDD_TARGET -> VTref	Target sense only, not a supply	3.3 V MCU target
CN4-2 SWCLK -> J-Link pin 4	Probe clock	3.3 V
CN4-3 GND -> pins 3/5	Signal return per probe instructions	Common reference
CN4-4 SWDIO -> pin 2	Bidirectional debug data	3.3 V
CN4-5 NRST -> pin 10	Active-low target reset; check SB12	3.3 V target
CN4-6 SWO -> pin 6	Optional trace output	Target dependent
Arty J10 USB-JTAG	Onboard bridge owns FPGA configuration	Board-powered; I/O standard per bank

The J-Link 9-pin 0.05-inch cable needs a verified keyed map to NUCLEO CN4; ST UM1724 directs removing CN2 jumpers for an external probe. VTref senses target voltage; it does not power the target. Use the Arty's own Digilent USB-JTAG bridge. Verify exact revision, connector orientation and I/O bank before any wiring. Never connect an incompatible GPIO voltage.

5. Backpropagation, verification and limitations

For x=1, y=2, h=tanh(wx+b), yhat=v*h+c and J=0.5*(yhat-y)^2, the initial values are h=0.4621171573, y_pred=0.4621171573, e=-1.5378828427 and J=1.182541819. The derivatives are dJ/dv=eh, dJ/dc=e, dJ/dw=ev(1-h^2)x and dJ/db=ev(1-h^2). Initial gradients (dw,db,dv,dc)=(-1.209464475,-1.209464475,-0.7106820475,-1.537882843). One simultaneous update at rate 0.1 gives (w,b,v,c) approx (0.620946448,0.120946448,1.071068205,0.153788284). The negative derivatives raise parameters and move the prediction toward the target. All four initial gradients passed central differences with epsilon 1e-6 and absolute error below 1e-5. After 100 further updates, loss was 1.70e-20. The candidate must handwrite and scan this derivation (maximum two pages) before submission.

Verification outcomes

CMake configure/build and the packaged launcher succeeded. PowerShell checks passed for the four expected mesh errors; finite-difference gradients; valid synthetic STL/CSV; rejection of empty, malformed/non-finite STL, wrong CSV header, NaN and a 20 MiB+1 CSV; and serial/parallel agreement. Browser inspection confirmed the local page, numerical result, batch table, device connection/telemetry and simulated timeout. Maximum accepted upload/import size is 20 MiB.

Limitations

No actual ANSYS export, physical hardware, firmware or bitstream was used. CSV example is synthetic; binary STL, native CAD solids and mesh registration are outside scope. The UI safely completes calculations but does not support cancellation. A clean Windows image and the VS Code debugger path were not independently tested. The packaged app was exercised on this development machine only. The report uses one observed benchmark set. Candidate must review the derivation and source, provide their actual candidate ID and handwrite/scan the two-page maximum appendix before submission.

AI-use and ownership

OpenAI Codex (GPT-6; exact runtime identifier not exposed) assisted with source and report drafting after the candidate asked it to review and complete the brief. Initial AI notes had incorrect gradient values; finite-difference checks exposed the error and they were corrected. The brief permits AI. The candidate should understand and revise this material and claim only work they can explain. Samples are self-created; no third-party implementation or dataset is bundled. No physical device behavior is claimed.

Hardware references: ST UM1724 · Digilent Arty manual · SEGGER 9-pin connector.